Do Defensive Equity Factors Deliver? Conditional Factor Premia over the U.S. Business Cycle, 1963–2026

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Abstract

A large practitioner literature markets "quality," profitability, and low-investment strategies as *defensive*: equity factors that protect capital when the economy contracts. We test this claim directly by measuring the realized premia of the five Fama–French factors and momentum conditional on the National Bureau of Economic Research (NBER) business-cycle state over 754 months from July 1963 to April 2026. The evidence is sharply heterogeneous. The market

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premium is strongly procyclical, falling by -16.4 percentage points per year in recessions; momentum is the most cyclical long-short factor, earning -7.7 points per year less in contractions and suffering its worst single month (-34.3% in April 2009). In contrast, the investment (CMA) and profitability (RMW) factors are genuinely defensive: their premia *rise* in recessions ($+6.6$ and $+1.1$ points per year, respectively), and CMA's annualized Sharpe ratio climbs from 0.31 in expansions to 1.01 in recessions. Size and value display no reliable cyclicality. The pattern survives replacement of the look-ahead NBER classification with a real-time, tradeable bear-market signal, and is visible in event time around both peaks and troughs. We conclude that "defensiveness" is a property of specific factors—those tied to firm fundamentals rather than to price trends—and not a generic feature of the factor zoo.

1 Introduction

The modern equity-factor industry is built on a simple promise: systematic exposures to characteristics such as value, size, profitability, investment, and momentum earn premia that compensate investors over the long run. Increasingly, that promise is accompanied by a second one. Profitability- and quality-tilted products in particular are marketed as *defensive*— strategies that not only earn a premium on average but also cushion portfolios precisely when investors value protection most, during economic downturns. Whether the cross-section of factor premia actually behaves this way over the business cycle is an empirical question with first-order implications for asset allocation, for risk-based explanations of the premia, and for the out-of-sample durability of the factors themselves.

The question matters because the leading economic interpretations of the factors make sharply different predictions about their cyclicality. Under a risk-based view, a positive unconditional premium must be compensation for losses in bad states (Cochrane, 2011). A factor that is truly risky should therefore *underperform* in recessions, when marginal utility is high; its premium is the reward for that procyclicality. Value has long been rationalized this way (Fama and French, 1993; Petkova and Zhang, 2005; Gulen et al., 2011), as has the aggregate market. Under a behavioral or limits-to-arbitrage view, by contrast, a factor's average return reflects mispricing that need not line up with the cycle at all, and a factor built on firm fundamentals could even act as a hedge if it loads on flight-to-quality demand. Momentum occupies a third category: its returns are known to be highly asymmetric, with infrequent but severe "crashes" concentrated in the rebounds that follow market troughs (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015). These competing views cannot all be right for all factors, and adjudicating among them requires measuring conditional, not just unconditional, performance.

This paper provides that measurement. We take the six workhorse U.S. equity factors—the market, size (SMB), value (HML), profitability (RMW), and investment (CMA) of Fama and French (2015), plus momentum (MOM) in the tradition of Jegadeesh and Titman (1993) and Carhart (1997)—and study how each factor's realized premium depends on the contemporaneous state of the U.S. business cycle as dated by the NBER. Our sample of 754 monthly observations spans July 1963 through April 2026 and contains 8 complete recessions accounting for 11.3% of all months. Throughout we report annualized premia, conduct inference with Newey and West (1987) heteroskedasticity- and autocorrelation-consistent standard errors, and complement the unconditional record with conditional Sharpe ratios, drawdowns, and event-time dynamics.

Three findings emerge. *First*, the market and momentum bear the business-cycle risk. The market earns -16.4 percentage points per year *less* in recessions than in expansions (a swing from a healthy expansion premium to an outright loss), and momentum earns -7.7 points per year less, turning a double-digit expansion premium into essentially zero. Momentum's recession Sharpe ratio collapses from 0.63 to 0.02, and its worst monthly return, -34.3%, lands in April 2009, in the violent recovery from the global financial crisis—the canonical momentum crash.

Second, and in direct contrast, the two fundamentals-based "quality" factors are genuinely defensive. The investment factor CMA earns $+6.6$ points per year *more* in recessions, and the profitability factor RMW earns $+1.1$ points more; both deliver their highest Sharpe ratios in contractions, with CMA's rising from 0.31 to a remarkable 1.01. These are precisely the exposures that practitioners label defensive, and the data vindicate the label. *Third*, size and value are acyclical: their recession-minus-expansion differentials are economically and statistically indistinguishable from zero, which is itself informative—value in particular does not display the procyclicality that a clean risk story would require over our full modern sample.

Because the NBER's business-cycle dates are announced with long lags and are therefore not investable, we verify that our conclusions do not depend on look-ahead information. We construct a real-time bear-market indicator that switches on whenever the trailing twelve-month total market return is negative, using only information available at the time. This simple, tradeable signal flags 21.3% of months and overlaps with 80% of NBER recession months, and it reproduces the cross-sectional pattern: momentum loses -11.1 points per year when the signal is on while RMW and CMA gain $+4.0$ and $+3.4$ points. The defensiveness of the quality factors is thus implementable, not an artifact of hindsight dating.

Our paper connects three literatures. The first is the asset-pricing work that defines and rationalizes the factors themselves (Fama and French, 1993, 2015; Novy-Marx, 2013; Cooper et al., 2008; Frazzini and Pedersen, 2014; Asness et al., 2019). We contribute a systematic, business-cycle-conditional accounting of these specific factors' realized performance. The second is the literature on time-varying risk premia and the link between expected returns and macroeconomic conditions (Chen et al., 1986; Fama and French, 1989; Lettau and Ludvigson, 2001; Henkel et al., 2011; Liew and Vassalou, 2000). Our event-time evidence—factor premia rising through contractions while the market falls into the trough and rebounds—is a compact, model-free picture of that time variation. The third is the work on momentum's crash risk (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015); we situate momentum at the cyclical extreme of the factor cross-section and show that its fragility is the mirror image of the quality factors' resilience. Finally, by documenting how much of each premium accrues in a small number of recession months, we speak to the fragility and out-of-sample durability of the factor zoo more broadly (McLean and Pontiff, 2016; Harvey et al., 2016).

The remainder of the paper proceeds as follows. Section 2 describes the data and the business-cycle classification. Section 3 establishes the unconditional benchmark. Section 4 presents the central conditional results. Section 5 replaces the NBER classification with a real-time signal. Section 6 examines event-time dynamics and momentum crashes, and Section 7 reports subsample robustness. Section 8 concludes.

2 Data and the Business-Cycle Classification

Factors. We obtain monthly returns on the five Fama–French factors and the momentum factor from Kenneth French's Data Library. The market factor (MKT) is the value-weighted return on all CRSP firms incorporated in the U.S. and listed on the NYSE, AMEX, or NASDAQ, in excess of the one-month Treasury bill rate. SMB (small minus big) and HML (high minus low book-to-market) are the size and value factors of Fama and French (1993); RMW (robust minus weak operating profitability) and CMA (conservative minus aggressive investment) are the profitability and investment factors of Fama and French (2015). MOM is the momentum factor (also known as WML or UMD) formed on prior 2–12 month returns. All factors are long-short, self-financing portfolios except the market, which is an excess return; we treat MKT as a tradeable factor throughout. Returns are in percent per month. Our merged sample is dictated by the availability of the five-factor data and runs from July 1963 to April 2026, a total of 754 months.

Business-cycle states. We classify each month as an expansion or a recession using the NBER's official business-cycle reference dates. Following standard practice, a month is coded as a recession if it falls strictly after a business-cycle peak and on or before the subsequent trough; under this convention the brief 2020 contraction spans March and April 2020. Our sample contains 8 complete recessions and 85 recession months, or 11.3% of the sample. The NBER dates are released with a substantial lag and are not knowable in real time; we treat them as the ex-post "ground truth" for the cycle and address implementability separately in Section 5.

Methods. For each factor we report the mean monthly return, its annualized counterpart ($12\times$ the monthly mean), annualized volatility ($\sqrt{12}\times$ the monthly standard deviation), and the annualized Sharpe ratio. Statistical inference on means and on conditional differences uses Newey and West (1987) HAC standard errors with the automatic lag selection $\lfloor 4(T/100)^{2/9} \rfloor$. Conditional means and their differences are estimated by regressing each factor return on a recession indicator, $r_{i,t} = \alpha_i + \beta_i \mathrm{REC}_t + \varepsilon_{i,t}$, so that α_i is the expansion mean and β_i is the recession-minus-expansion differential. Maximum drawdown is computed from the cumulative return path. All results are reproducible from the accompanying R pipeline.

3 Unconditional Factor Performance

Table 1 establishes the unconditional benchmark. Over the full sample every factor earns a positive average premium, and all but SMB are individually significant at conventional levels. The market commands the largest annualized premium, followed closely by momentum, whose annualized

Table 1: Unconditional factor performance, July 1963–April 2026. Monthly and annualized mean premia (%), annualized volatility (%), annualized Sharpe ratio, the Newey–West t-statistic for the mean, skewness, excess kurtosis, maximum drawdown (%), and the share of months with positive returns.

Factor	Mean (mo.)	Mean (ann.)	Vol. (ann.)	Sharpe	t(mean)	Skew	Kurt.	Max DD	% Pos.
MKT	0.60	7.17	15.47	0.46	3.59	-0.49	1.70	-55.8	59.9
SMB	0.19	2.22	10.46	0.21	1.58	0.36	3.06	-56.5	50.5
HML	0.30	3.56	10.29	0.35	2.29	0.09	2.16	-57.8	53.6
RMW	0.26	3.09	7.71	0.40	2.88	-0.33	10.86	-41.8	56.1
CMA	0.24	2.92	7.17	0.41	2.84	0.27	1.34	-27.3	52.3
MOM	0.61	7.31	14.47	0.51	3.95	-1.29	9.68	-57.8	61.8

Sharpe ratio of 0.51 is the highest of the six factors. The quality factors RMW and CMA earn smaller premia but at notably lower volatility, so their Sharpe ratios (0.40 and 0.41) rival HML's and exceed SMB's. Size is the weakest performer, with a Sharpe ratio of only 0.21.

These unconditional moments, however, conceal the feature that motivates our study: the shape of the return distribution differs starkly across factors. Momentum combines its high mean with pronounced negative skewness and heavy tails, and the largest maximum drawdown of any long-short factor—the statistical signature of crash risk emphasized by Daniel and Moskowitz (2016). The quality factors, by contrast, exhibit the mildest drawdowns. A mean-variance investor looking only at Table 1 would over-weight momentum and under-appreciate exactly when its losses arrive. The conditional analysis that follows makes that timing explicit.

4 Factor Premia over the Business Cycle

Table 2 is the heart of the paper. For each factor it reports the annualized premium in expansions and in recessions, their difference with a Newey–West t-statistic, the Sharpe ratio in each state, and the worst recession-month outcome. The cross-section splits cleanly into three groups.

The cyclical factors: market and momentum. The market premium swings from a robust $+9.0\%$ per year in expansions to -7.4% in recessions, a differential of -16.4 points ($t = -1.73$). This is the textbook procyclicality of equity risk and the benchmark against which factor "defensiveness" should be judged. Momentum is the most cyclical of the long-short factors: its premium falls by -7.7 points per year in recessions ($t = -0.88$), collapsing from a double-digit expansion premium to near zero, and its conditional Sharpe ratio falls from 0.63 to essentially 0.02. Momentum also posts the worst single recession-month return in the table. A strategy that looks defensive on average is, in fact, maximally exposed to the cycle.

The defensive factors: profitability and investment. The investment and profitability factors behave in the opposite way. CMA earns +2.2% per year in expansions but 8.7% in recessions, a differential of +6.6 points, and its Sharpe ratio *rises* from 0.31 to 1.01. RMW similarly improves, from +3.0% to 4.1% per year (+1.1 points), and both factors exhibit their smallest drawdowns during contractions. While the individual differentials are not estimated with high precision—recessions are rare, so conditional means are noisy—their *sign* is economically meaningful and consistent: the factors that the industry markets as defensive do, in this sample, earn more when the economy contracts, not less. This is the central result of the paper.

The acyclical factors: size and value. Size and value display essentially no cyclicality. HML's recession-minus-expansion differential is −0.0 points and SMB's is −0.2 points; neither is distinguishable from zero. The HML result is notable given the long tradition of interpreting value as compensation for distress risk that bites in bad times (Petkova and Zhang, 2005; Gulen et al., 2011): over our full modern sample, value does not reliably underperform in recessions. We return to the stability of these estimates in Section 7.

Figure 1 visualizes the table. The contrast between the dark recession bars for MKT and MOM (which fall sharply) and those for RMW and CMA (which rise) is the paper's main message in a single picture.

Figure 2 places the conditional results in historical context, plotting the cumulative value of $1 invested in each factor on a log scale with NBER recessions shaded. The market and momentum series visibly stall or reverse inside the shaded bands, whereas the RMW and CMA series tend to grind upward through them. Momentum's long, smooth ascent punctuated by abrupt cliffs—most

Table 2: Factor performance conditional on the NBER business-cycle state. Annualized mean premia (%) in expansions and recessions, their difference (recession minus expansion) with the Newey–West t-statistic in parentheses, the annualized Sharpe ratio in each state, the maximum drawdown during recession months (%), and the worst single recession-month return (%). *, **, *** denote significance at the 10%, 5%, and 1% levels.

	Mean return (ann. %)				Sharpe		Rec.	Worst
Factor	Expansion	Recession	Diff.	t(Diff.)	Exp.	Rec.	MaxDD	month
MKT	9.01	-7.37	-16.38*	(-1.73)	0.64	-0.31	-59.7	-17.2
SMB	2.25	2.01	-0.24	(-0.06)	0.22	0.15	-20.6	-8.2
HML	3.56	3.55	-0.00	(-0.00)	0.36	0.26	-30.7	-13.8
RMW	2.96	4.06	1.10	(0.32)	0.38	0.52	-12.2	-4.3
CMA	2.18	8.74	6.56	(1.54)	0.31	1.01	-14.6	-6.0
MOM	8.19	0.46	-7.72	(-0.88)	0.63	0.02	-51.0	-34.3

dramatically in 2009—is the cumulative footprint of the crash risk documented below.

5 A Real-Time, Implementable Signal

A natural objection to Section 4 is that the NBER's business-cycle dates are not known in real time: the committee announces peaks and troughs with lags that can exceed a year. The conditional premia in Table 2 are therefore informative about the *nature* of the factors but are not by themselves a tradeable strategy. To show that the defensive pattern does not hinge on look-ahead information, we replace the NBER classification with a simple signal constructed only from past prices.

We define a month as a "bear" month if the trailing twelve-month total return on the market (through the end of the prior month) is negative, and a "bull" month otherwise. This rule uses only information available before the return is earned and could have been implemented in real time throughout the sample. It classifies 21.3% of months as bear and captures 80% of all NBER recession months, so it is a noisy but economically sensible proxy for bad times.

Table 3 repeats the conditional analysis using this signal. The cross-sectional pattern is preserved. Momentum earns -11.1 points per year less in bear months ($t = -1.75$), the largest and most significant cyclical exposure in the table, while RMW and CMA again *gain* (+4.0 and +3.4 points per year, respectively). The market premium is lower in bear months, as expected. That a tradeable, real-time signal recovers the same ordering as the ex-post NBER dates is reassuring: the

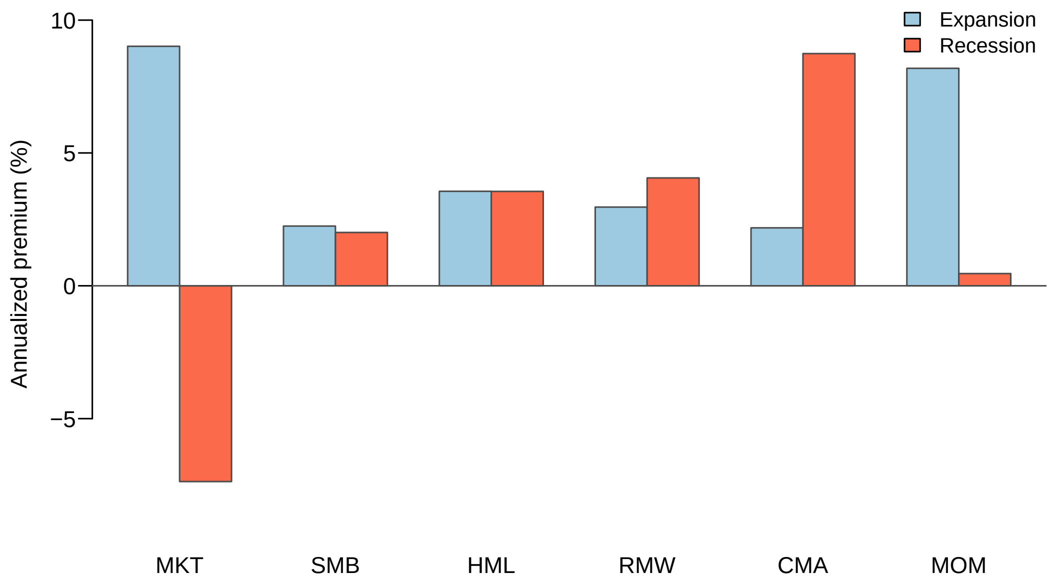

Figure 1: Annualized factor premia by business-cycle state. Each pair of bars reports a factor's annualized mean premium in NBER expansions (light) and recessions (dark). The market and momentum premia fall sharply in recessions; the profitability (RMW) and investment (CMA) premia rise.

defensiveness of the quality factors is a feature an investor could actually have exploited, not a product of hindsight.

6 Event-Time Dynamics and Momentum Crashes

The conditional means in Table 2 average over the entire recession state. To see *when* within the cycle each factor earns or loses, we construct event studies around the 8 NBER peaks and troughs in our sample. For each event we accumulate factor returns over a window of twelve months on either side and average across episodes. Figure 3 reports the results.

The left panel, centered on peaks, shows the market rolling over almost exactly at the peak and declining steadily thereafter, the visual definition of procyclical risk. RMW and CMA, by contrast, continue to accumulate returns through and beyond the peak. The right panel, centered on troughs, is even more striking. In the year before a trough the market falls by roughly fifteen percent on average and then rebounds sharply once the trough passes. The quality factors climb monotonically

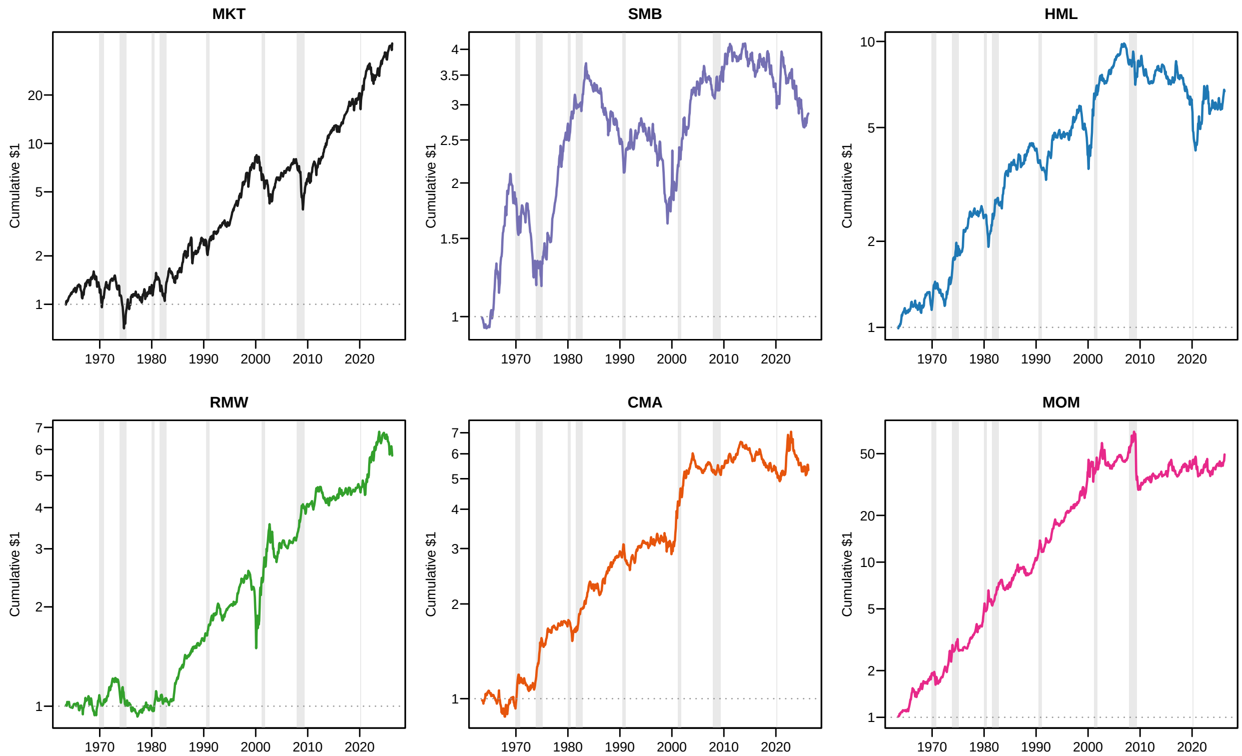

Figure 2: Cumulative factor performance with recessions shaded. Cumulative value of $1 invested in each long-short factor (the market is an excess return), plotted on a logarithmic scale. Shaded vertical bands are NBER recessions. Note the log scale differs across panels.

throughout, providing their steadiest returns exactly when the market is in free fall—the defensive profile in motion.

Momentum tells the cautionary tale. It performs well *into* downturns, riding the prevailing trend, but stalls and reverses in the immediate aftermath of the trough, when the market's violent rebound whipsaws portfolios that are short the prior losers. This is the Daniel and Moskowitz (2016) crash mechanism. The single worst momentum month in our sample, −34.3%, occurs in April 2009, during the rebound from the financial-crisis trough. More systematically, momentum earns only 1.7% per year (annualized) in the twelve months following NBER troughs, against 8.2% in all other months—the premium all but vanishes precisely in the recovery window where its crashes cluster. An investor who held momentum for its high unconditional Sharpe ratio would have absorbed these losses at the worst possible time, a risk that the quality factors do not carry.

Table 3: Factor performance under a real-time bear-market signal. A month is classified "bear" if the trailing twelve-month total market return (through the prior month) is negative, using only ex-ante information. The table reports annualized mean premia (%) in bull and bear months, their difference with the Newey–West t-statistic in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

	Mean return (ann. %)			
Factor	Bull	Bear	Diff.	t(Diff.)
MKT	8.32	2.34	-5.98	(-0.92)
SMB	1.64	4.97	3.33	(1.03)
HML	3.20	4.23	1.03	(0.30)
RMW	2.29	6.33	4.04	(1.44)
CMA	2.20	5.58	3.38	(1.18)
MOM	9.62	-1.45	-11.07*	(-1.75)

7 Robustness: Subsample Stability

Because recessions are infrequent, conditional means are estimated from a limited number of months and could be driven by a single episode. As a check on stability, Table 4 re-estimates the recession-minus-expansion differential separately in the first and second halves of the sample (split at 1990). The qualitative ordering is robust: the market's differential is large and negative in both halves, and momentum's is negative in both. The defensiveness of the quality factors is present in both subsamples but rotates across them—CMA's positive differential is concentrated in the earlier period, while RMW's is strongest after 1990—consistent with the two factors capturing related but distinct facets of firm quality. Value's differential is positive and significant before 1990 but turns negative afterward, which is why its full-sample cyclicality washes out; this instability is itself a useful caution against strong risk-based claims for HML over the modern sample.

We caution that the conditional differentials are estimated with limited precision and that our classification is binary; richer conditioning on the *depth* of contractions, or on continuous macroeconomic state variables in the spirit of Lettau and Ludvigson (2001) and Henkel et al. (2011), is a natural extension. The robustness of the *sign* pattern across samples, across the NBER and real-time classifications, and in event time nonetheless gives us confidence in the paper's qualitative conclusions.

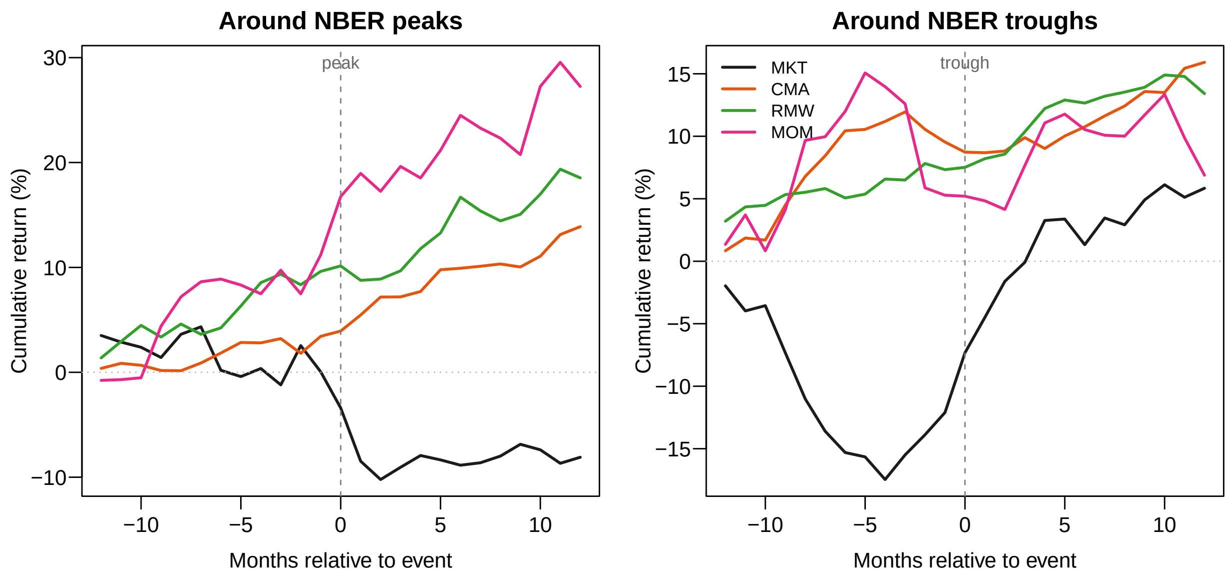

Figure 3: Event-time cumulative factor returns around NBER peaks and troughs. Each line is the average cumulative return (%) of a factor in event time, accumulated from twelve months before the event and averaged across the 8 in-sample episodes. The vertical dashed line marks the business-cycle turning point. Left: peaks. Right: troughs.

8 Conclusion

We have asked whether the equity factors marketed as defensive actually deliver when the economy contracts, and we find that the answer depends entirely on which factor one means. The market and momentum are procyclical: they earn their premia in good times and surrender them—momentum violently—in bad times. The profitability and investment factors are genuinely defensive: their premia rise in recessions, their Sharpe ratios peak in contractions, and they accumulate returns steadily while the market falls into and climbs out of its troughs. Size and value are acyclical, and value's lack of reliable procyclicality over the modern sample sits awkwardly with risk-based interpretations. These conclusions hold under the ex-post NBER dating, under a real-time tradeable signal, and in event time.

The practical implication is that "defensiveness" is not a generic property of factor investing but a specific property of factors built on firm fundamentals—profitability and conservative investment—as opposed to those built on prices and trends. For asset allocation, this suggests that quality tilts can play the diversifying role often claimed for them, whereas momentum, despite its high unconditional

Table 4: Subsample stability of the recession differential. Annualized recession-minus-expansion differential (%) for each factor, estimated separately over 1963–1989 and 1990–2026, with Newey–West t-statistics in parentheses.

	1963–1989		1990–2026	
Factor	Diff.	t	Diff.	t
MKT	-11.18	(-0.91)	-21.73	(-1.37)
SMB	-4.30	(-0.73)	3.65	(0.56)
HML	8.37	(1.44)	-12.05	(-1.55)
RMW	-4.31	(-1.20)	8.73	(1.84)
CMA	13.87	(2.69)	-3.34	(-0.67)
MOM	-2.41	(-0.34)	-16.24	(-0.87)

Sharpe ratio, concentrates its losses in exactly the states where diversification is most valuable. For asset-pricing theory, the heterogeneity we document is a reminder that a positive average premium is consistent with very different conditional risk profiles, and that distinguishing among the factors requires looking at when, not just whether, they pay.

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